

⑧ Applications of ML / Real Life ML

B2B

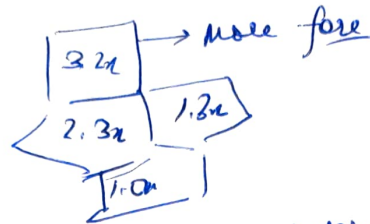
1. Retail - Amazon / Big Bazaar

Targeted Marketing: - ~~to all~~

Association rule Based learning → Baby diapers & Beer.

② Banking & Finance

③ Transport - Ola



4. Manufacturing - Tata → Predictive maintenance

5. Consumer internet →

Sentiment Analysis → Movie ~~review~~ reviews examples

E.g. # tweet → Collect all tweets
60% N/A → 40% MB

Great data to check

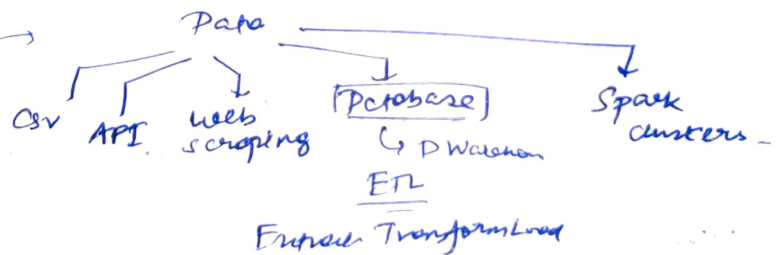
Send the data to Store Breaking companies.

⑨ Machine Learning Development Life Cycle (MLDLC)

SDLC → Software development Life Cycle.

(i) Frame the Problem

(ii) Gathering the Data →



(iii) Data Preprocessing → Remove duplicates

- Missing val
- Outliers
- Scale / Standardise

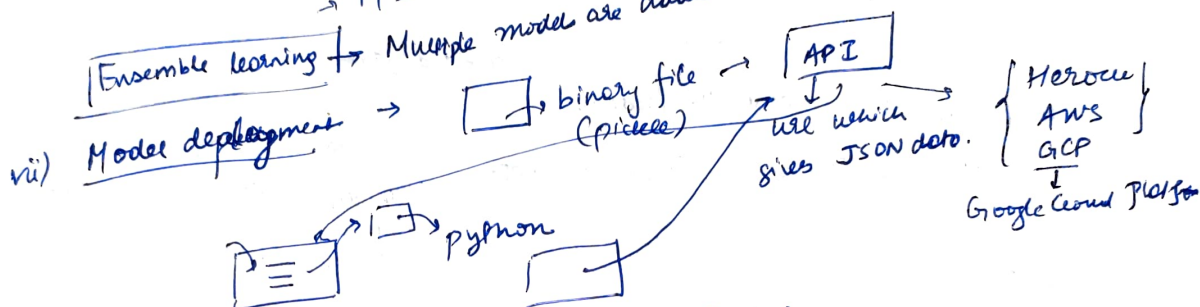
- iv) Exploratory Data Analysis
 Experiments with data
 i) Visualization, (ii) Univariate/Bivariate, (iii) Outlier detection.
 iv) Imbalanced data set.

v) Feature Engineering and Selection

→ Intelligent changes in existing features, Feature Selection.

- vi) Model Training, Evaluation & Selection
 ↓
 Different models → Accuracy
 → MSE
 Evaluate diff. models
 1 or multiple Algorithms.
 Hyperparameter tuning.

Ensemble learning → Multiple models are added to make new model (powerful)



- viii) Testing / Beta Testing →
 Trusted customers give feedback.
A/B testing

ix) Optimize

- Backup
- Data
- Model
- Load Balancing

Retrain otherwise rotting

⑩ Data Engineer vs DA vs DS vs ML Engineer

• Analyst → Startup Posing.
 Plan → Data → Processing → EPA → Feature Engg → Training → Deploy → Test → Optimize

(i) Data Engineer →

OLTP → Online Transaction Processing. (~~Data~~ Data Base)
 ↓
 OLAP → Online Analytical Processing (Data warehouse)
 ↓
 pipelines

Skills → i) PSA

ii) Java/R/Python

iii) Advanced DBMS

iv) Bigdata Tools. (Apache Spark, Hadoop, Apache Kafka, Apache Hive)

v) Cloud Platforms (AWS, GCP)

vi) Distributed Systems

vii) Data pipelines.

(i) Data Analyst

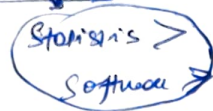
- Clean & organize raw data
- Data visualization
- Collaborating
- Data collection procedures optimization
- Producing & maintaining reports

Skills

- Statistics
- Business Acumen — Medium to High
- SQL
- Data Story Telling
- Data Mining, Cleaning & Merging
- Advanced MS Excel

(ii) Data Scientist (Full stack)

(Decision Scientist)



SDE
statistician

→ Data ko smjh krke future prediction krne.

(iii) ML Engineer

- Deploying ML models
- Monitoring & maintenance
- Scaling & optimizing the model for prediction

Skills

- Maths
- PL
- Distributed Systems

• SE & System design

11. What are Tensors? Tensor in-depth

DATA SCIENTIST

Analytical skills
Business Acumen
Data Storytelling

Soft Skills

(i) Containers for nos.

(ii) 0D Tensor / Scalars →

(2) (3)

Scalars →

0D Tensor

(iii) $[1, 2, 3, 4] \sim$ 1D Tensor

vector nDim → 1

4D

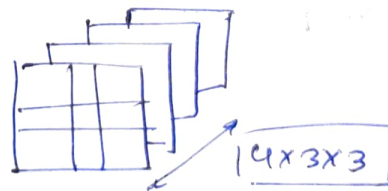
Axis → 2dim

No. of Axis = rank = dimension

(iv) 2D Tensor / Matrices

$\begin{bmatrix} [1, 2, 3], \\ [4, 5, 1], \\ [7, 8, 9] \end{bmatrix} \leftarrow 2D$

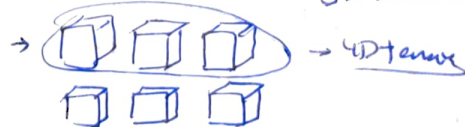
(v) ND Tensors



3D Tensor

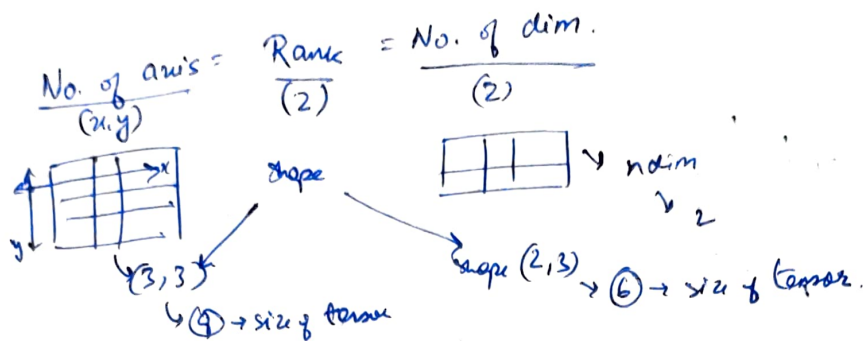
(vi)

5D Tensor



0 — 5D Tensor

vi. Rank, Axes and Shape



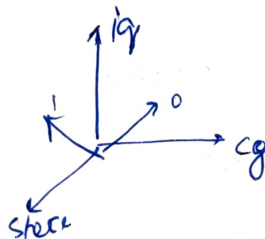
vii) Example of 1D Tensor

Students $\rightarrow$

CG	iq	state	placement
8.1	91	WB	4

WB=0, KR=1

$[8.1, 91, 0] - 1D$
 (3D dimension of vector)



viii) Example of 2D Tensor

2D tensor
 $\begin{bmatrix} [\dots] \\ [\dots] \end{bmatrix} \rightarrow \text{vector} - 1D$
 $\rightarrow [D]$

CG iq state
 $[D] = \begin{bmatrix} [\dots] \\ [\dots] \\ [\dots] \end{bmatrix}$ 2D Tensor
 $\rightarrow$ 1D Tensor

ix) Example of 3D Tensor - used in Natural Language Processing.

[NLP] $\rightarrow$ Hi Nitish
 Hi Rahul
 Hi Ankit

vectorisation
 Hi | Nitish | Rahul | Ankit
 1 0 0 0
 0 1 0 0
 0 0 1 0
 0 0 0 1

$(3, 2, 4)$
 shape $\Rightarrow 3 \times 2 \times 4 = 24$

Time series data

Store $\rightarrow$ highest | lowest

Day1 $\rightarrow$ $\rightarrow 2D \rightarrow 10$

Day2

...

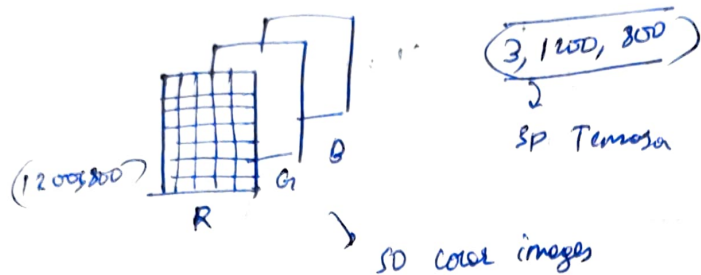
Day 365

(365, 2)

$(10 \times 365 \times 2)$
 $\downarrow$
 time axis

x) Example of 4D Tensors

images $\rightarrow$ Computer Vision



3 channels (R, G, B), 1200, 800

(50, 3, 1200, 800) $\rightarrow$ 4D Tensor

DL - CN

xii) Example of 5D Tensors

60 sec video $\rightarrow$

$\hookrightarrow$ 30fps

$\hookrightarrow$ 480p $\rightarrow$ 480 x 720

(4, (1800, 480, 720, 3))

$\uparrow$ 5D tensor

float 32 $\rightarrow$ 28476.56 MB

(27 GB)

$\hookrightarrow$ video

$\hookrightarrow$ mpeg $\rightarrow$ mp4

(12) 3. Virtual Environment

base (root)

Deployment $\rightarrow$ you have deploy all libraries in server

Create new environment

i) Safe box

ii) Using Kaggle